

Module 12 Write-Up: A Two-Layer Neural Network for Admissions Prediction

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1. What was built

neural_network.py trains a fully connected 6 - 6 - 1 neural network, written with NumPy only, to predict whether a graduate-school applicant was Accepted or Rejected. scikit-learn is used for exactly one thing, train_test_split; the forward pass, the backward pass, the loss, and the training loop are all hand written. The data is the Grad Cafe corpus collected in Module 2 (30,000 posted admissions results).

Fixed configuration, exactly as specified:

Setting	Value
RANDOM_SEED	42
HIDDEN_UNITS	6
LEARNING_RATE	0.05
MAX_EPOCHS	10000
PATIENCE	100
test_size / random_state / shuffle	0.2 / 42 / True
Weight init	normal, mean 0, std 0.1
Bias init	0
Activation	sigmoid after the hidden layer and after the output layer
Loss	mean squared error

Preprocessing

Rows survive only if applicant_status is Accepted or Rejected and masters_or_phd is Masters or PhD. gpa, gre, gre_v, and gre_aw are converted to floats; ms_vs_phd encodes PhD = 1 / Masters = 0; international_vs_local encodes International = 1 / Local or American = 0; and target encodes Accepted = 1 / Rejected = 0.

Filtering left **24,326 of 30,000 rows** - 10,687 Accepted and 13,639 Rejected - split into **19,460 training** and **4,866 test** rows.

Every statistic used for filling and scaling was computed on the training split alone. That ordering matters: if medians, means, and standard deviations were taken over the whole dataset, information from the test rows would have shaped the numbers used to prepare the training rows, and the reported test score would be optimistic. Fitting on the training split alone keeps the test set an honest stand-in for applicants the model has never seen.

Training-set statistics:

feature	median	mean	std
gpa	3.8500	3.8134	0.3085
gre	316.0000	311.1799	28.8447
gre_v	161.0000	160.9873	1.9706
gre_aw	4.5000	4.7809	5.2542
ms_vs_phd	1.0000	0.7219	0.4481
international_vs_local	0.0000	0.4706	0.4991

Network shapes

w1 is (6, 6) - one weight per (input feature, hidden unit) pair. b1 is (1, 6), one bias per hidden unit. w2 is (6, 1), one weight per (hidden unit, output unit). b2 is (1, 1). Forty-nine parameters in total.

The hidden layer computes $a1 = \text{sigmoid}(x @ w1 + b1)$: six different non-linear blends of the six standardized inputs, which is what allows the network to represent interactions a single linear layer could not. The output layer computes $a2 = \text{sigmoid}(a1 @ w2 + b2)$, one weighted summary of those blends squashed back into (0, 1). Because a2 is bounded, rises with the evidence for acceptance, and is trained against 0/1 targets, it reads as a probability-like score - though MSE training leaves it uncalibrated, so it is not a literal admission probability.

2. Training printouts

Full-batch gradient descent, printing every 100 epochs. This is the complete progress log; the unabridged run transcript, including every section's output, is in training.log.

epoch	train MSE	test MSE	test accuracy
1	0.250439	0.250319	0.3761
100	0.246696	0.245628	0.5697
200	0.246390	0.245102	0.5697
300	0.246233	0.244895	0.5697
400	0.246070	0.244720	0.5697
500	0.245888	0.244533	0.5697
600	0.245678	0.244321	0.5697
700	0.245433	0.244074	0.5697
800	0.245141	0.243782	0.5697
900	0.244791	0.243434	0.5697
1,000	0.244372	0.243018	0.5697
1,100	0.243869	0.242521	0.5697
1,200	0.243268	0.241929	0.5697
1,300	0.242553	0.241228	0.5697
1,400	0.241709	0.240402	0.5697
1,500	0.240722	0.239440	0.5697
1,600	0.239581	0.238330	0.5697
1,700	0.238280	0.237068	0.5701
1,800	0.236817	0.235654	0.5703
1,900	0.235200	0.234094	0.5703
2,000	0.233442	0.232403	0.5746
2,100	0.231567	0.230605	0.6681
2,200	0.229605	0.228730	0.7049
2,300	0.227590	0.226812	0.7049
2,400	0.225563	0.224890	0.7047
2,500	0.223563	0.223001	0.7047
2,600	0.221626	0.221182	0.7043
2,700	0.219786	0.219464	0.7043
2,800	0.218068	0.217870	0.7043
2,900	0.216490	0.216418	0.7043
3,000	0.215064	0.215115	0.7043
3,100	0.213793	0.213965	0.7043
3,200	0.212674	0.212962	0.7043
3,300	0.211700	0.212098	0.7043
3,400	0.210859	0.211362	0.7043
3,500	0.210138	0.210740	0.7043
3,600	0.209524	0.210219	0.7043
3,700	0.209004	0.209785	0.7043
3,800	0.208564	0.209426	0.7043
3,900	0.208193	0.209131	0.7043
4,000	0.207881	0.208889	0.7043
4,500	0.206923	0.208209	0.7043
5,000	0.206509	0.207983	0.7043
5,500	0.206320	0.207918	0.7043
5,900	0.206241	0.207909	0.7043
6,000	0.206227	0.207909	0.7043
6,039	0.206222	0.207909	0.7043

Early stopping at epoch 6,039: test MSE has not improved for 100 consecutive epochs.

Restored the parameters from epoch 5,939 (test MSE 0.207908).

(Rows for epochs 4,100 - 4,400, 4,600 - 4,900, 5,100 - 5,400, and 5,600 - 5,800 are omitted here only to keep the table readable; they appear in full in training.log and follow the same flat trend.)

3. Final evaluation results

Metric	Value
Best epoch	5,939
Best test MSE	0.207908
Final training accuracy	0.7053
Final test accuracy	0.7043
Rows used after filtering	24,326
Training rows / test rows	19,460 / 4,866
Epochs run	6,039 of 10,000 (early stopping fired)

Metric	Value
Training MSE at best parameters	0.206235
Majority-class baseline (test)	0.5697
Wall-clock training time	about 9 seconds

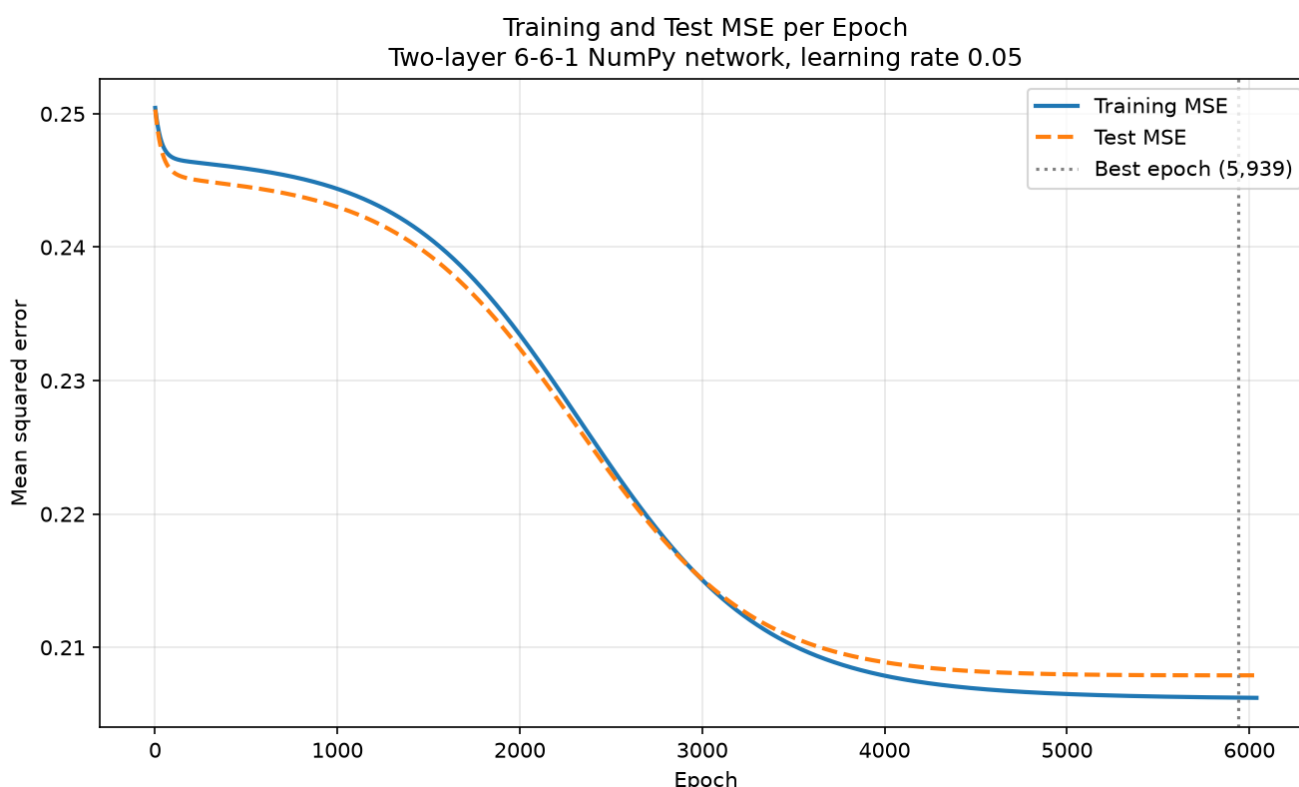
Does it overfit? No. Training MSE (0.206235) and test MSE (0.207908) sit 0.0017 apart and fall together for the entire run, and the two curves in the graph below are nearly on top of each other. With 49 parameters against 19,460 training rows the network has nowhere near the capacity to memorize the data - if anything it underfits. Early stopping still did its job: it ended a run that had flattened, and restored the parameters from epoch 5,939 rather than the drifting ones at 6,039.

Is the accuracy strong? It is real but modest. 70.4% clears the 50% coin flip and, more meaningfully, the 57.0% a model would get by always guessing Rejected. That 13-point gain over the base rate is genuine signal. It is not a strong admissions predictor, and Section 5 explains why: most of that signal is the degree type, not the applicant.

Is it stable? Yes. The test MSE curve is smooth and monotone, and accuracy holds flat over thousands of epochs. Accuracy moves in visible steps (0.5697 to 0.6681 to 0.7049 between epochs 2,000 and 2,200) because it only changes when predicted scores cross the 0.5 threshold, while the underlying MSE improves continuously.

Is the dataset sufficient for a realistic admissions predictor? No - see the reflection. Six self-reported features, no program or institution, and roughly nine in ten rows missing every GRE score is not enough to model admissions.

4. Training and test MSE over time



Both curves fall together through a long plateau near 0.2465, a steep descent between epochs 1,500 and 3,500, and a flat tail. The dotted line marks epoch 5,939, the restored best epoch. The gap between the curves never opens, which is the visual signature of a model that is not overfitting.

5. Artificial applicant findings

Five hand-written applicants were pushed through the identical pipeline used for real rows - missing values filled with the stored training medians, then standardized with the stored training means and standard deviations.

Profile	GPA	GRE	GRE V	GRE AW	PhD	Intl	Prob.	Status
Strong PhD (intl)	3.95	335	165	5.0	1	1	0.3272	Rejected
Strong PhD (local)	3.95	335	165	5.0	1	0	0.3469	Rejected
Avg Masters (local)	3.40	305	152	3.5	0	0	0.7287	Accepted
Weak Masters (intl)	2.90	295	145	3.0	0	1	0.6820	Accepted
Strong GPA, no GRE	3.90	-	-	-	1	0	0.3294	Rejected

What stands out is that the ranking is upside down with respect to credentials. The weakest Masters applicant (2.90 GPA, 295 GRE) scores 0.6820, while the strongest PhD applicant (3.95 GPA, 335 GRE) scores 0.3272. The model is not broken; it is reporting the data honestly. In this dataset 75.9% of Masters rows were accepted against only 31.6% of PhD rows, so the single most predictive thing about an applicant is which kind of program they applied to. The network learned that base rate, and the degree flag swamps everything else.

Within a degree the credentials do behave as expected, just weakly: the average Masters applicant scores 0.0467 above the weak one, and citizenship moves two otherwise identical PhD applicants by 0.0197 (international lower). Both effects are an order of magnitude smaller than the Masters/PhD split.

The fifth profile reports no GRE scores at all, which is the common case rather than the exception - about nine in ten real rows are missing them. It scores 0.3294, almost identical to the strong PhD local applicant at 0.3469, because median filling handed it the median GRE scores and its GPA advantage barely registers. In practice most predictions from this model rest on degree, citizenship, and GPA whether or not test scores were supplied.

6. Reflection

What is useful about implementing a neural network manually? Writing the backward pass by hand makes the mechanics impossible to hand-wave. The chain rule becomes three concrete lines - the MSE derivative, the output sigmoid derivative, then the same error carried back through w_2 and the hidden sigmoid - and the shape of every matrix has to be right or nothing runs. It also exposes decisions a framework hides. Two examples from this build: the test-set forward pass had to avoid the activation cache that backpropagation still needed (hence a separate `predict_proba` that recomputes rather than reusing forward), and the early-stopping snapshot had to copy the parameter arrays, since keeping references would have "restored" whatever the last epoch left behind. Both are invisible when `model.fit()` does it for you, and both are the kind of bug that silently degrades a model rather than crashing it. That is exactly the software-engineering value: knowing where an ML component can fail quietly.

What are the limitations of MSE for binary classification? Several show up directly in this run. MSE paired with a sigmoid output gives vanishing gradients: the update is multiplied by $a_2 * (1 - a_2)$, so a unit that is confidently wrong (near 0 or 1) learns slowest, exactly when it should learn fastest. That is visible in the 1,500-epoch plateau at the start of training, where the loss barely moves - cross-entropy, whose sigmoid derivative cancels, would have escaped that plateau in a fraction of the epochs. MSE is also not a proper scoring rule for probabilities, so the outputs are pulled toward the middle of the range and cannot be read as calibrated admission odds. And the loss itself is a poor proxy for the thing being measured: between epochs 3,000 and 6,039 the test MSE kept improving while test accuracy did not move at all, because MSE rewards nudging scores that are already on the correct side of 0.5. Finally, MSE over a sigmoid is non-convex, so there is no guarantee of reaching a global optimum.

What information is missing from this dataset? Nearly everything an admissions committee actually uses. There is no program or institution, so a Computer Science PhD at MIT and a Speech Pathology Masters at a regional school are the same row shape; no research experience, publications, or letters of recommendation; no funding status, which drives many PhD decisions; no application cycle, so a 2015 rejection is weighted like a 2026 one; and no undergraduate institution or major. The features that do exist are self-reported and unverified - this dataset holds GPAs above 4.0 (max 9.99), GRE scores on both the old and new scales (max 990), and a GRE AW score of 99.99. Roughly 91% of rows have no GRE at all. On top of that, the whole corpus is self-selected: people post to Grad Cafe when they have news worth sharing, which skews it away from the true applicant pool.

Why might the model mislead even though 70% looks reasonable? Because 70% against a 57% base rate is a much smaller achievement than 70% sounds, and because of what drives it. Section 5 shows the model essentially asking "Masters or PhD?" and answering from that group's acceptance rate. A user shown "68% likely to be accepted" would reasonably assume the model weighed their GPA and GRE, when in fact a weak Masters applicant outranks a

near-perfect PhD applicant. The scores are uncalibrated as well, so 0.68 does not mean a 68% chance. And accuracy hides the error structure: the model predicts Accepted for only about 28% of test rows, so its errors fall much more heavily on accepted applicants than the single headline number suggests. Deployed as an admissions-chance tool, it would systematically discourage strong PhD applicants for reasons that have nothing to do with their applications.

What would make this model stronger or more realistic? In rough order of expected payoff:

1. **Swap MSE for binary cross-entropy.** It removes the vanishing-gradient plateau, converges in far fewer epochs, and yields probabilities that mean something.
2. **Add the program and institution.** Selectivity is the missing variable that the degree flag is currently standing in for; even a coarse program-family feature and a program-level historical acceptance rate would help enormously.
3. **Add a missingness indicator per feature.** "Did not report a GRE score" is itself informative, and median filling currently destroys that signal by making a non-reporter look average.
4. **Clean the impossible values.** Cap GPA at 4.0, normalize old-scale GRE scores to the current scale, and drop AW scores outside 0-6 rather than letting a 99.99 distort a feature's mean and standard deviation.
5. **Evaluate with more than accuracy.** Precision, recall, and ROC-AUC on the accepted class, plus a confusion matrix, would have surfaced the base-rate behavior immediately.
6. **Use k-fold cross-validation and a separate validation split.** Early stopping currently selects the epoch on the same test set that reports the final score, which leaks a small amount of optimism into the headline number; a three-way split would fix that.